FINAL-YEAR COMPUTER ENGINEERING · SIIT, THAMMASAT UNIVERSITY

Singha Junchan

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I build access control for cloud and AI systems.

Open to: security internships from January 2027, in Thailand or abroad · full-time roles from May 2027

COMPETITIVE PROGRAMMING
#1 in Thailand
TOI18 national champion, 2022, after 1,000+ hours of training

CLIENT PROJECT
TU Alert
built for **Thammasat University**; **tech lead** (architecture, security, code); security built in; launching soon

RESEARCH
9 research projects
2 accepted (one **Best Paper**), 3 submitted, 4 in preparation; first author on 8

INDUSTRY
2 internships
Allianz Ayudhya (security governance and risk, with the CISO) and **Amity Solutions** (LLM chatbots)

LEADERSHIP AND TEACHING
11 student events
led SIIT First Meet 2024 (about 500 people) and organised 10 more; founded **two coaching platforms**

CERTIFICATION
ISC2 Certified
Certified in Cybersecurity (CC), 2025; CompTIA Security+ planned Dec 2026

01 Experience

knowledge | tech used

Tech Lead and Developer, TU Alert Thammasat University (client project) 2026 — Launching soon

Campus incident reporting **inside LINE**, built for Thammasat University (launching soon). **Tech lead**: I lead the team, design the architecture and the security, and write code.

LINE Official Account, group bot and two LIFF apps: students report with photo and map pin, track and chat; staff accept, transfer and resolve from a phone or the portal. **Security built in**: roles, second-step sign-in, audit log, PDPA erasure.

System architecture · Secure web development · TypeScript · Next.js

Information Security Intern Allianz Ayudhya Jun to Jul 2026

Learned **security governance and risk** from the CISO, and helped design how the team's **third-party risk** work uses AI.
Built a **questionnaire agent** (evidence, a reason and a **confidence score** per answer; **gap and fit-gap analysis**) and a **guidance agent** that checks inputs, suggests next steps and flags risks. Tested on real, anonymized cases.

Information security governance · Risk management · Third-party risk management · Due-diligence questionnaires

Founder, Curriculum Author and Lead Instructor TOI Tutor and ABIG Center (online) 2022 to present

Founded two olympiad coaching platforms (2022); coached hundreds of students.

Algorithms and data structures · Competitive programming · C++ · Python

R&D Intern, AI research team Amity Solutions Jul to Aug 2023

Read **LLM research papers**, applied their ideas and **tested them end to end**.

Large language models · Retrieval-augmented generation (RAG) · Applying research papers · End-to-end testing

02 Research

9 research projects: 2 accepted, 3 submitted, 4 in preparation. First author on 8.

FEDAC 1st of 4 authors — Under review · IEEE TDSC

Checks access **once**, then gives a **short-lived pass** per service; sign-in gets stronger for sensitive services, and the rules can be checked on a blockchain.

Zero trust · Zero-knowledge proofs · Service mesh · Blockchain audit · Cloud benchmarking · Go · Rust · Groth16 zk-SNARK · Envoy + Istio · IPFS · AWS EKS

U-HAP 2nd of 4 authors · presented the paper — Presented · IEEE JCSSE 2026 · Best Paper Award

All Kubernetes access-rule types in **one graph**, prepared ahead of time so each request is a **quick lookup**; a deny always wins.

Access control (RBAC, ABAC, ACL) · Kubernetes admission · Policy compilation · Graph models · Python · Kubernetes webhook · OPA (baseline)

ZKAudit First author — Accepted · KST 2027

An auditor **proves it may audit without revealing its credential**, and Merkle chains show that the logs were not changed.

Zero-knowledge proofs · Cloud audit logs · Merkle commitments · Privacy · Rust · Groth16 zk-SNARKs · AWS EC2

Post-quantum authorization across Kubernetes clusters First author — Submitted

Each cluster **checks for itself** that an access decision used every allow and deny rule, with **post-quantum** signatures (ML-DSA, ML-KEM).

Post-quantum cryptography · Multi-cluster Kubernetes · Authorization · Distributed ledgers · Go · ML-DSA · ML-KEM · Hyperledger Fabric · AWS EKS

Also submitted: Self-healing fog computing for industrial IoT | **Also in preparation:** MoCA: AI containment · ZK-ALBA · Threshold-gated encrypted SIEM · Fresh answers from elastic multi-cloud services

03 Projects

Novel Nest reading and writing platform (Next.js, MySQL); **project manager and product owner**, team of four

Automated Cat Feeder IoT feeder with web control and live camera; **MQTT pipeline**; Bronze Medal, SCIUS 2022

Arduino retro console handheld game console in C++; **I wrote the menu, controls and drawing loop**

Velour Mall UX/UI case study in Figma: indoor navigation, parking assistant, AI chatbot

04 Education

B.Eng. in Computer Engineering

Aug 2023 to May 2027 (expected)

Minor in Cloud Computing and Cybersecurity. Sirindhorn International Institute of Technology (SIIT), Thammasat University.

GPA 3.67 / 4.00, first-class honours expected. Full scholarship: 4-Year Continuing Academic Outstanding Student Program.

05 Awards

2022 **National Champion: 1st place in Thailand and Gold Medal** Thailand Olympiad in Informatics (TOI18)

2026 **Best Paper Award** IEEE JCSSE 2026, Bangkok, for U-HAP (2nd of 4 authors)

2025 **1st place overall** Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT

REFERENCES Contact details on request. **Apichai Wongchindakhun** Chief Information Security Officer (CISO), Allianz Ayudhya
Prof. Somchart Fugkeaw Professor, School of ICT, SIIT, Thammasat University **Pachara** Co-developer, TU Alert, SIIT, Thammasat University